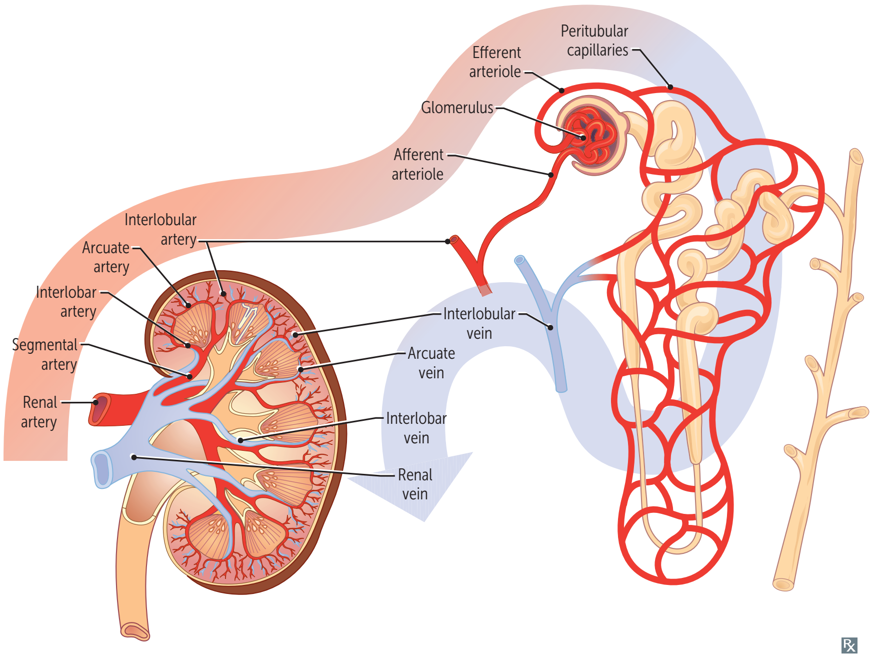
Vesicoureteral reflux

Retrograde flow of urine from bladder toward upper urinary tract. Can be 1° due to abnormal/insufficient insertion of the ureter within the vesicular wall (ureterovesical junction [UVJ]) or 2° due to abnormally high bladder pressure resulting in retrograde flow via the UVJ. ↑ risk of recurrent UTIs.

▸ RENAL—ANATOMY

Renal blood flow

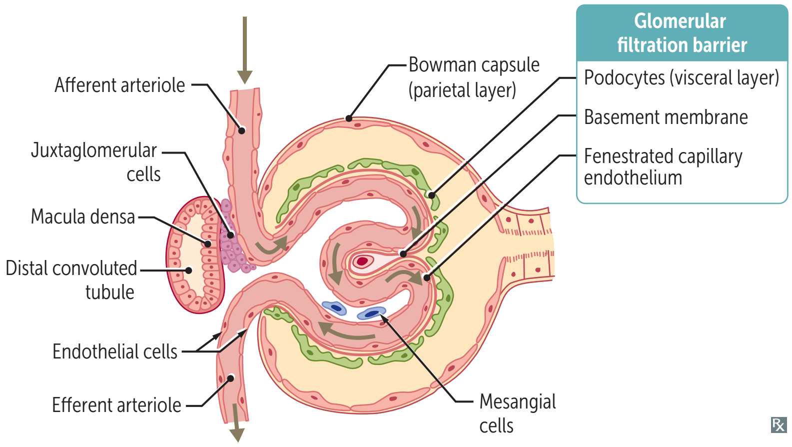


Left renal vein receives two additional veins: left suprarenal and left gonadal veins.

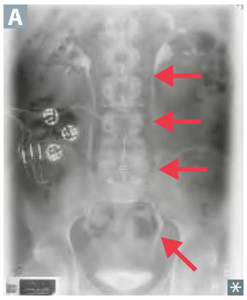
Renal medulla receives significantly less blood flow than the renal cortex. This makes medulla very sensitive to hypoxia and vulnerable to ischemic damage (eg, ATN).

Left kidney is most commonly taken during **L**iving donor transp**L**antation because it has a **L**onger renal vein (rule of **L**s).

Glomerular anatomy



Course of ureters



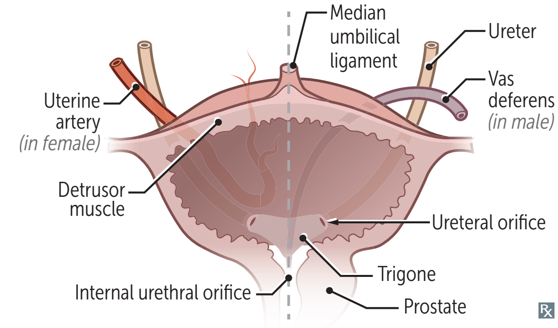
Course of ureter A: arises from renal pelvis, travels under gonadal arteries → over common iliac artery → under uterine artery/vas deferens (retroperitoneal).

Gynecologic procedures (eg, ligation of uterine or ovarian vessels) may damage ureter → ureteral obstruction or leak.

Bladder contraction compresses the intramural ureter, preventing urine reflux.

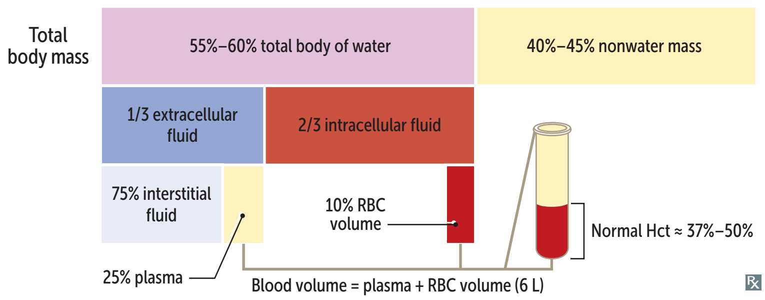
3 common points of ureteral obstruction: ureteropelvic junction, pelvic inlet, ureterovesical junction.

Water (ureters) flows **over** the iliacs and **under** the bridge (uterine artery or vas deferens).



▸ RENAL—PHYSIOLOGY

Fluid compartments



HIKIN': **HI K^+ IN**tracellularly.

60–40–20 rule (% of body weight for average person):
- 60% total body water
- 40% ICF, mainly composed of K^+, Mg^{2+}, organic phosphates (eg, ATP)
- 20% ECF, mainly composed of Na^+, Cl^-, HCO_3^-, albumin

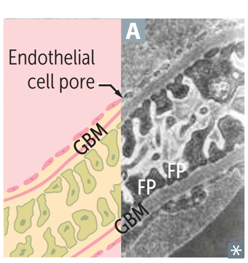
Plasma volume can be measured by radiolabeling albumin.

Extracellular volume can be measured by inulin or mannitol.

Serum osmolality = 275–295 mOsm/kg H_2O.

Plasma volume = TBV × (1 – Hct).

Glomerular filtration barrier



Responsible for filtration of plasma according to size and charge selectivity.

Composed of
- Fenestrated capillary endothelium
- Glomerular basement membrane (GBM) with type IV collagen chains and heparan sulfate
- Visceral epithelial layer consisting of podocyte foot processes (FPs) A

Charge barrier—glomerular filtration barrier contains ⊖ charged glycoproteins that prevent entry of ⊖ charged molecules (eg, albumin).

Size barrier—fenestrated capillary endothelium (prevents entry of > 100 nm molecules/blood cells); slit diaphragm (prevents entry of molecules > 40–50 nm).

Podocyte foot processes interpose with GBM—both charge and size barrier.